

**CSCE 211 Assignment 1**

1. Convert the following as indicated: (0.4 points)

(a)  $00100101_2$  to a decimal integer.

$$(1 * 2^0) + (0 * 2^1) + (1 * 2^2) + (0 * 2^3) + (0 * 2^4) + (1 * 2^5) + (0 * 2^6) + (0 * 2^7)$$

Decimal = 37

(b)  $00101001_2$  to a hexadecimal integer.

$$0010 = (0 * 2^0) + (1 * 2^1) + (0 * 2^2) + (0 * 2^3) = 2$$

$$1001 = (1 * 2^0) + (0 * 2^1) + (0 * 2^2) + (1 * 2^3) = 9$$

Hexadecimal = 29

(c)  $B_{16}$  to an 8-bit unsigned binary integer.

B=11

$$11 \% 2 = 1; 5 \% 2 = 1; 2 \% 2 = 0; 1 \% 2 = 1$$

B=1011

1=1

$$1 \% 2 = 1; 0 \% 2 = 0; 0 \% 2 = 0; 0 \% 2 = 0$$

1=0001

B1=1011 0001

(d)  $212_{10}$  to an 8-bit unsigned binary integer.

$$212 \% 2 = 0; 106 \% 2 = 0; 53 \% 2 = 1; 26 \% 2 = 0; 13 \% 2 = 1; 6 \% 2 = 0; 3 \% 2 = 1; 1 \% 2 = 1$$

Binary = 11010100

2. Compute the 4-bit binary sum of the following 4-bit unsigned binary integers. Provide the base-10 result as well. Do allow values to overflow—that is do not add bits in excess of the 4 bits. Additionally, provide decimal(base<sub>10</sub>) integer values: (0.4 points)

(a) 0101 + 1001

$$\begin{array}{r} 0101 \\ + 1001 \\ \hline 1110 \end{array} \rightarrow \text{Decimal} = (0 * 2^0) + (1 * 2^1) + (1 * 2^2) + (1 * 2^3) = 14$$

$$0101 \rightarrow (1 * 2^0) + (0 * 2^1) + (1 * 2^2) + (0 * 2^3) = 5$$

$$1001 \rightarrow (1 * 2^0) + (0 * 2^1) + (0 * 2^2) + (1 * 2^3) = 9$$

$$5 + 9 = 14$$

(b) 1010 + 0110

$$\begin{array}{r} 1010 \\ + 0110 \\ \hline 0000 \end{array} \rightarrow \text{Decimal} = (0 * 2^0) + (0 * 2^1) + (0 * 2^2) + (0 * 2^3) = 0$$

$$1010 \rightarrow (0 * 2^0) + (1 * 2^1) + (0 * 2^2) + (1 * 2^3) = 10$$

$$0110 \rightarrow (0 * 2^0) + (1 * 2^1) + (1 * 2^2) + (0 * 2^3) = 6$$

$$6 + 10 = 16 \text{ but with overflow, the answer is } 0$$

3. Convert the following to 8-bit signed binary integers and perform the indicated operations: (0.4 points)

(a)  $1F_{16} - 20_{16}$

Converting Hexadecimal to Binary

$1=1$

$1\%2=1; 0\%2=0; 0\%2=0; 0\%2=0$

$1=0001$

$F=15$

$15\%2=1; 7\%2=1; 3\%2=1; 1\%2=1$

$15=1111$

$1F=0001\ 1111$

$2=2$

$2\%2=0; 1\%2=1; 0\%2=0; 0\%2=0$

$2=0010$

$0=0$

$0\%2=0; 0\%2=0; 0\%2=0; 0\%2=0$

$0=0000$

$20=0010\ 0000$

Because 00100000 is negative we must use Two's Complement:

(Swap)  $00100000=11011111$

(Add 1)  $11011111+1=11100000$

Now we can add the two 8-bit Signed integers

$00011111$

$+ 11100000$  (Signed Binary Answer)

$11111111$

Because the first bit is 1, the answer is negative, so we must use Two's Complement:

(Swap)  $11111111=00000000$

(Add 1)  $00000000+1=00000001$

So, the answer is -00000001 in binary or -1 in decimal.

(b)  $31_{10} - 17_{10}$

$31\%2=1; 15\%2=1; 7\%2=1; 3\%2=1; 1\%2=1; 0\%2=0; 0\%2=0; 0\%2=0$   
 $31=00011111$

$17\%2=1; 8\%2=0; 4\%2=0; 2\%2=0; 1\%2=1; 0\%2=0; 0\%2=0; 0\%2=0$   
 $17=00010001$

Because 00010001 is negative we must use Two's Complement:

(Swap)  $00010001=11101110$

(Add 1)  $11101110+1=11101111$

Now we can add the two 8-bit Signed integers

$11101111$   
 $+ 00011111$  (Signed Binary Answer)  
 $00001110$

So the answer is 00001110 in binary or 14 in decimal.

4. For each of the following, show their conversion to binary coded decimals (BCD) as 8421-code:  
(0.4 points)

(a)  $127_{10}$

Convert each digit to 4-bit binary OR you can use Table 1.7 in the textbook.

1=0001

2=0010

7=0111

The answer is 000100100111

(b)  $F7_{16}$

Convert to decimal:

$15 \cdot 16 + 7 = 247$

Convert each digit to 4-bit binary OR you can use Table 1.7 in the textbook.

2=0010

4=0100

7=0111

The answer is 001001000111

5. Decode the two following 8-bit binary strings into ASCII characters: (0.4 points)

(a) 01100011 01010011 01000011 01100101

After subtracting the first zero from each string of characters, we can use Table 1.8 in the Textbook

110 0011= c

101 0011= S

100 0011= C

110 0101= e

The text spells cSCe

(b) 01110100 01101100 00111011 01100100 01110010

111 0100= t

110 1100= l

011 1011= ;

110 0100= d

111 0010= r

The text spells tl;dr